Jin Hyun Park

Ph.D. Student, Computer Science and Engineering, Texas A&M University

jinhyun.park@tamu.edu jinhyunpark.com Google Scholar 7201krap jinhyunjasonpark

Research Interests

Machine learning, computational neuroscience, and natural language processing. In particular, biologically inspired neural network architectures, the evolutionary origins of predictive behavior, and learning-based control for physical systems.

Education

Ph.D. in Computer Science and Engineering Aug 2022 – May 2027 (expected)
Texas A&M University, College Station, TX, USA Advisor: Yoonsuck Choe

B.Sc. in Computer Science, First Class Honours Sep 2018 – Jul 2021
University of Bristol, Bristol, UK Advisors: Rui Ponte Costa, Cian O'Donnell

Publications

C: conference, J: journal, W: workshop, A: arXiv preprint. * denotes equal contribution.

[C2] Soha Lee, Soojin Lee, Heesung Yang, Hyunju Song, Hyunji Lee, Jinsan An, Jeongwan Shin, **Jin Hyun Park**, Jun Lee, Hyeyoung Park, Kilim Nam. *KoNeoBench: A Curated Evaluation Dataset for LLM Understanding of Korean Neologisms*. Findings of EMNLP, 2026. [arXiv:2609.19916]

[C1] Seyyed Ali Ayati*, **Jin Hyun Park***, Yichen Cai, Marcus Botacin. *Making Acoustic Side-Channel Attacks on Noisy Keyboards Viable with LLM-Assisted Spectrograms' "Typo" Correction*. 19th USENIX Workshop on Offensive Technologies (WOOT), 2025. [arXiv:2504.11622]

[J1] William Kang, Chris Anand, **Jin Hyun Park**, Yoonsuck Choe. *Evolutionary Factors Contributing to the Emergence of Prediction*. Adaptive Behavior, 2025. [doi:10.1177/10597123251364746]

[W1] Alex Alvarez*, **Jin Hyun Park***, Yoonsuck Choe. *Saliency Thresholds in Neural Code and its Relation to the Power-Law, Gaussian, and Lambert W Function*. NeurIPS Workshop on Symmetry and Geometry in Neural Representations (NeurReps), 2025. [OpenReview]

[A2] **Jin Hyun Park**, Cheng Zhang, Yoonsuck Choe. *Incorporating Visual Cortical Lateral Connection Properties into CNN: Recurrent Activation and Excitatory-Inhibitory Separation*. arXiv preprint, 2025. [arXiv:2509.15460]

[A1] **Jin Hyun Park**. *Representations Learnt by SGD and Adaptive Learning Rules: Conditions that Vary Sparsity and Selectivity in Neural Networks*. arXiv preprint, 2022. [arXiv:2201.11653]

Posters

[P2] *Predictable Internal Dynamics Mark Robust Recurrent Reinforcement-Learning Agents*. The 26th KOCSEA Technical Symposium, 2026.

[P1] *Incorporating Visual Cortical Lateral Connection Properties into CNN: Recurrent Activation and Excitatory-Inhibitory Separation*. Annual Meeting of the Korean Society for Computational Neuroscience, 2026.

Research Experience

Graduate Research Assistant Aug 2022 – Present
Brain Networks Lab, Texas A&M University Advisor: Yoonsuck Choe

- Biologically inspired neural networks: incorporated visual-cortex lateral connectivity (recurrent activation, excitatory–inhibitory separation) into CNNs [A2] and analyzed saliency thresholds in neural codes [W1].
- Evolution of prediction: studied the evolutionary factors that give rise to predictive behavior in simulated agents [J1].
- Learning-based control: developed an ML-based smart thermostat for ondol floor heating with Hansung (Korean heating company), predicting room temperature response from sensor time series.

Research Assistant Jan 2022 – Jul 2022
Multimodal Language and Cognition Lab, Kyungpook National University, Daegu, Korea

- Research on classical NLP algorithms and large language models.
- Organized a reinforcement learning seminar series for lab members, with lecture notes and code samples.

Industry Experience

Physical AI Ph.D. Fellow Feb 2026 – Present
Doosan Corporation, Seoul, Korea

Machine Learning Engineer Intern May – Aug 2023; Jul – Aug 2025
Machine Learning Team, Tilda Corp., Seoul, Korea
- 2025: Meta-heuristic and neural methods for vehicle routing, multi-agent path finding, and time-series forecasting.
- 2023: Machine learning solutions based on a neural-network-guided genetic algorithm.

Machine Learning Engineer Intern Jul – Aug 2024
AI Service Part, Digital Innovation Team, Samsung Life Insurance, Seoul, Korea
- Built an automatic PDF parsing pipeline and a retrieval-augmented (RAG) search engine for insurance documents and policy retrieval.

Machine Learning Engineer Intern May – Jul 2024
AI Prediction Tech Team, Data Science Lab, LG Uplus, Seoul, Korea
- Ensemble and transformer models for identifying outbound roaming travelers.
- Large-scale data generation, preprocessing, and visualization with SQL and Google BigQuery.

Business Strategy Intern Oct – Dec 2021
Commercial Vehicle Business Innovation Team, Hyundai Motor Company, Seoul, Korea
- Short- to long-term business modeling for commercial vehicles, with a focus on FCEV and BEV.

Teaching and Service

Reviewer / Program Committee
AAAI 2024, 2025, 2026; AAAI-AIA 2026; AAAI-Demo 2026; IEEE Transactions on Cognitive and Developmental Systems 2025; NeurIPS NeurReps 2026.

Teaching Assistant 2022 – Present
CSCE 420/625 Artificial Intelligence, Texas A&M University.

Seminar 2025
Seminar on large language models, Yonsei University.

Selected Projects

Depth- and Attention-Aware RGB Image Inpainting 2025
Course project, Computational Photography [arXiv:2505.00735]
- Dual-encoder attention architecture fusing RGB and depth features; outperforms RGB-only baselines on several metrics, with Grad-CAM confirming improved spatial reasoning.

PEPSE: Personal Photo Search Engine 2024
Course project, Information Storage and Retrieval [demo video] [project page]
- Semantic photo search combining caption generation, query embedding, and face recognition with pre-trained transformer models.

Fellowships, Grants, and Awards

Physical AI Ph.D. Fellowship, Doosan Corporation 2026 – Present
Academic Excellence Grant, Texas A&M University 2025 – 2027
Travel Grants
- Doosan Corporation, Consumer Electronics Show (CES), 2026
- Texas A&M University, NeurIPS, 2025
- Samsung Life Insurance and LG Uplus, internship travel, 2024
- Hyundai Mobis, Consumer Electronics Show (CES), 2024

Barry Thomas Scholarship, University of Bristol 2018
Future Prize Winner Scholarship, University of Bristol 2017

Skills

Languages	Python, Java, C/C++, SQL
Frameworks	PyTorch, OpenCV, Google BigQuery
Areas	Deep learning, computational neuroscience, reinforcement learning, NLP and LLMs, retrieval-augmented generation, evolutionary and meta-heuristic optimization